

RS003 Week 3 — English Worksheet

Topic: Build the Ship Body · **Course:** Pygame Space Shooter · **Time:** about 45 minutes

This week you think about the code that draws the first piece of your spaceship: a **body** that sits in the middle-bottom of the screen. You read code that draws shapes, stores important numbers as **constants**, and finds the centre of the screen with **integer division** (`//`). You will read it, fix it, and explain it.

Words to know: **constant**, **rectangle**, **circle**, **integer division** (`//`), **centre**

1 · Predict 🧠

Read each piece of code. Just from reading it, write what you think it does. Do not run it.

```
WIDTH = 800
CENTER_X = WIDTH // 2
```

What number is `CENTER_X` ? Why use `//` instead of `/` here?


```
pygame.draw.rect(screen, CYAN, (CENTER_X - 20, HEIGHT - 80, 40, 50))
```

The four numbers are (left, top, width, height). Where on the screen does this rectangle appear, and how big is it?


```
pygame.draw.circle(screen, WHITE, (CENTER_X, HEIGHT - 90), 8)
```

What shape, where, and how big? What does the last number 8 control?


```
screen.fill(SPACE)
pygame.draw.rect(screen, CYAN, (CENTER_X - 20, HEIGHT - 80, 40, 50))
pygame.display.flip()
```

In what order do these three lines run? What would you see if `screen.fill(SPACE)` came AFTER the rectangle instead of before it?

2 · Spot the Bug 🐛

Each block below was meant to do something but is broken. Fix it on paper, then explain why the original was wrong.

Bug A — The ship body should sit in the **centre** across the screen. Right now it is glued to the left edge.

```
pygame.draw.rect(screen, CYAN, (0, HEIGHT - 80, 40, 50))
```

Hint: use `CENTER_X` to place it in the middle.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — `CENTER_X` is supposed to be the middle of the screen, but it comes out as a number with a decimal point and the drawing looks off.

```
CENTER_X = WIDTH / 2
```

Hint: screen positions must be whole numbers.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The body is supposed to sit near the **bottom**, where the player ship belongs. Right now it is drawn near the top of the screen.

```
pygame.draw.rect(screen, CYAN, (CENTER_X - 20, 80, 40, 50))
```

Hint: in Pygame, Y = 0 is the **top**. The bottom is near **HEIGHT**.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Explain the Code

Read this working example line by line. Then answer the questions below by reading only — you do not need to run it.

```
import pygame
pygame.init()

WIDTH = 800
HEIGHT = 600
CENTER_X = WIDTH // 2

screen = pygame.display.set_mode((WIDTH, HEIGHT))
pygame.display.set_caption("Space Shooter")

SPACE = (10, 12, 40)
CYAN = (80, 220, 255)

clock = pygame.time.Clock()
running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    screen.fill(SPACE)
    pygame.draw.rect(screen, CYAN, (CENTER_X - 20, HEIGHT - 80, 40, 50))
    pygame.display.flip()
    clock.tick(60)

pygame.quit()
```

WIDTH, **HEIGHT**, **SPACE**, and **CYAN** are all constants. What is each one used for?

The rectangle's left value is `CENTER_X - 20` , not `CENTER_X` . Why subtract 20? What would happen if you used `CENTER_X` on its own?

What does the `while running:` loop do? What makes the loop stop?

Why is the cyan rectangle drawn in the middle across the screen? Point to the exact lines that make this happen.

Is the rectangle near the top or near the bottom of the window? How can you tell from the numbers?

4 · Explain Your Lesson Code 🎥

In today's live lesson you wrote code with your teacher to draw your ship body. Now explain that code in a short phone video. You may show it running on the screen. Try to use these words: **constant, rectangle, circle, integer division, centre.**

1. Show your ship body sitting in the middle-bottom of the screen.
2. Read your `CENTER_X` line out loud and say what `//` does.
3. Point at your rectangle's four numbers and name each one (left, top, width, height).
4. Say why `Y = 0` is the top, not the bottom.

Write what you will say in your video. Plan it here before you record.

Submit ☒

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.